SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	JAGADISH.V	**Reg. No.:**	192521352
Section / Batch:		**Date:**	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Q57. Application: Disaster Response Communication System

A disaster response system processes communication requests using the recurrence $T(n) = 3T(n/2) + n \log n$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how the system performs as the communication load increases. Interpret efficiency using asymptotic notation.

Q58. Application: Autonomous Drone Coordination System

An autonomous drone coordination system manages flight data using the recurrence $T(n) = 2T(n/3) + n^2$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how coordination performance changes with increasing drone count. Interpret efficiency using asymptotic notation.

Code of Conduct

I JAGADISH.V(192521352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V JAGADISH

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Q58

Introduction

An Autonomous Drone Coordination System manages communication, navigation, collision avoidance, and task allocation among multiple drones. To improve efficiency, the system divides the drone network into smaller groups and recursively processes their flight information. The recurrence relation describing the running time is:

$$T(n) = 2T(n/3) + n^2$$

where:

- n = Number of drones
- 2T(n/3) = Two recursive subproblems, each handling one-third of the drones
- n^2 = Time required for coordination, communication, collision checking, and synchronization at the current level

Step 1: Identify the Master Theorem Parameters

The standard Master Theorem form is:

$$T(n) = aT(n/b) + f(n)$$

Comparing with the given recurrence:

$$T(n) = 2T(n/3) + n^2$$

we get:

- a = 2
- b = 3
- f(n)=n^2

Step 2: Calculate $n^{\log_b a}$

Using the formula:

$$n^{\log_b a}$$

Substitute the values:

$$n^{\log_3 2}$$

Since

$$\log_3 2 \approx 0.631$$

Therefore,

$$n^{\log_3 2} = n^{0.631}$$

Step 3: Compare $f(n)$with $n^{\log_b a}$

Given:

$$f(n) = n^2$$

and

$$n^{\log_3 2} = n^{0.631}$$

Clearly,

$$n^2 \gg n^{0.631}$$

Hence,

$$f(n) = \Omega(n^{0.631+\varepsilon})$$

for some positive constant ε.

Step 4: Verify the Regularity Conditio

The regularity condition is:

$$af(n/b) \leq cf(n)$$

where $c < 1$.

Substitute the values:

$$2\left(\frac{n}{3}\right)^2$$
$$= \frac{2}{9}n^2$$

Since

$$\frac{2}{9} < 1$$

the condition is satisfied.

Step 5: Select the Master Theorem Case
Since

- $f(n)$grows faster than $n^{\log_b a}$, and
- the regularity condition is satisfied,

the recurrence falls under Master Theorem Case 3.

Step 6: Time Complexity
Therefore,

$$\boxed{T(n) = \Theta(n^2)}$$

The overall running time of the autonomous drone coordination system is quadratic.

Recursion Tree Interpretation
At each recursive level:

- The problem is divided into 2 smaller groups.
- Each group contains n/3 drones.
- The coordination work at the current level is n^2.

The work decreases rapidly at lower levels:

- Level 0 → n^2
- Level 1 → $\frac{2}{9}n^2$
- Level 2 → $\left(\frac{2}{9}\right)^2 n^2$
- …

The total work forms a geometric series dominated by the first term, giving:

$$T(n) = \Theta(n^2)$$

Q57

Introduction

A **Disaster Response Communication System** manages emergency communication requests during disasters such as earthquakes, floods, cyclones, or fires. The system divides incoming requests into smaller groups, processes them simultaneously, and combines the results efficiently.

The recurrence relation representing the processing time is:

$$T(n) = 3T(n/2) + n\log n$$

where:

- **n** = Number of communication requests.
- **3T(n/2)** = Three subproblems, each handling half of the requests.
- **n log n** = Time required to merge, prioritize, and coordinate communication data.

Step 1: Identify the Master Theorem Parameters

The standard Master Theorem form is:

$$T(n) = aT(n/b) + f(n)$$

Comparing with the given recurrence,

$$T(n) = 3T(n/2) + n\log n$$

we identify:

- **a = 3**
- **b = 2**
- **f(n)=n\log n**

Step 2: Calculate $n^{\log_b a}$

$$n^{\log_2 3}$$

Since,

$$\log_2 3 \approx 1.585$$

Therefore,

$$n^{\log_2 3} = n^{1.585}$$

Step 3: Compare $f(n)$with $n^{\log_b a}$

Given,

$$f(n) = n\log n$$

Compare:

- $n\log n$
- $n^{1.585}$

Since

$$n\log n = O(n^{1.585-\epsilon})$$

for some constant $\epsilon > 0$,

the function $f(n)$grows **slower** than $n^{1.585}$.

Step 4: Select the Correct Master Theorem Case

The Master Theorem states:

Case 1

If

$$f(n) = O\left(n^{\log_b a - \epsilon}\right)$$

then

$$T(n) = \Theta\left(n^{\log_b a}\right)$$

Since

$$n \log n = O(n^{1.585-\epsilon})$$

Step 5: Final Time Complexity

Therefore,

$$T(n) = \Theta\left(n^{\log_2 3}\right)$$

or

$$\boxed{T(n) = \Theta(n^{1.585})}$$

Conclusion

The recurrence relation

$$T(n) = 3T(n/2) + n \log n$$

matches the Master Theorem with:

- $a = 3$
- $b = 2$
- $f(n) = n \log n$

Since $n \log n$grows asymptotically slower than $n^{\log_2 3}$, **Master Theorem Case 1** applies.

Hence, the overall time complexity is:

$$\boxed{T(n) = \Theta\left(n^{\log_2 3}\right) = \Theta(n^{1.585})}$$